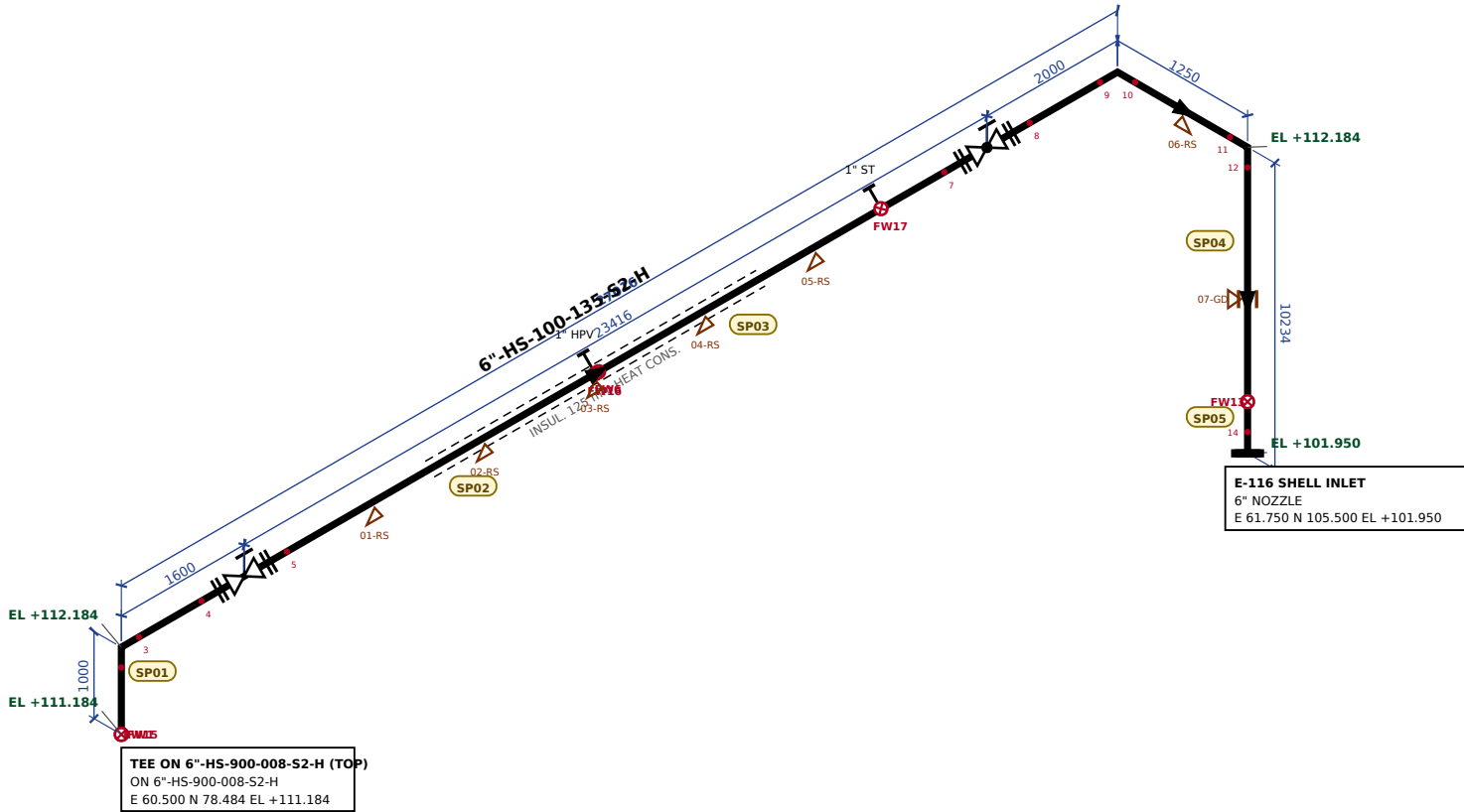
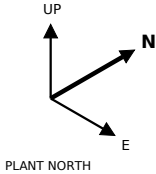


ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	PIPE SCH 40, BE, ASTM A335 P11 SMLS	6"	36.4 m
2	ELBOW 90 LR, BW, SCH 40, ASTM A234 WP11	6"	3
3	FLANGE WN RF CL600 B16.5, bore SCH 40, ASTM A182 F11	6"	5
4	SOCKOLET 3000# (branch), ASTM A182 F11	1"	2
5	ERECTION: GATE VALVE OS&Y CL600 RF, A217 WC6, trim 8	6"	1
6	GLOBE VALVE CL600 RF, A217 WC6, trim 8	6"	1
7	GASKET SPW 304/graphite, outer ring, CL600 B16.20	6"	5
8	STUD BOLT SET (nuts) A193 B7 / A194 2H, CL600	6"	5
9	PIPE SUPPORTS (std. dwgs): guide x1, rest x6	-	7

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	6"	771	SP01-2	6"	975	SP02-1	6"	11650
SP03-1	6"	10973	SP04-1	6"	1375	SP04-2	6"	793
SP04-3	6"	8750	SP05-1	6"	1138			

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=2.3m SP02=11.8m SP03=11.1m SP04=11.9m SP05=1.3m

NOTES

- Pipe/fittings/flanges per class S2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 81.1 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES (P-No.4) 705-745 C. NDE: 100 % RT of BW; 100 % MT/PT of fillets; HB <= 225; 100 % PMI.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Branch taken from top of HP header; line falls to E-116 - drains into shell (no pocket).
- S2 (P11) - PWHT and 100 % RT; hot bolting check at 430 C.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT		
PIPING ISOMETRIC 6"-HS-100-135-S2-H HP STEAM HEADER TO E-116		
DISCIPLINE PIPING	STAGE FEED	SCALE NTS
SHEET 1 OF 1		
DRAWN CLAUDE CODE	CHECKED -	APPROVED -
DATE 2026-10-02		
DRAWING NUMBER CFU-100-PI-ISO-012		REV A

LINE NUMBER 6"-HS-100-135-S2-H	FROM HP steam header	TO E-116	PIPING CLASS S2 - 600# 1.25Cr-1/2Mo ASTM A335 P11
DESIGN P / T 46.0 barg / 430 C	OPER. P / T 41.4 barg / 400 C	TEST P (HYDRO) 81.1 barg (1.5xPxST/S=1.18)	PIPE WALL SCH 40 (7.11 mm)
INSULATION H 125 mm	PWHT YES (P-No.4) 705-745 C	NDE 100 % RT of BW; 100 % MT/PT of fillets;	STRESS CAT 1 - T > 300 C & NPS >= 6; 600# class & T > 200
P&ID CFU-100-PR-PID-010	FLUID / PHASE HS / V HP steam to stabiliser r	PAINT HT silicone/Al (insul.)	SPOOLS / WELDS 5 spools; 11 shop / 6 field